COMS 4720/5720 Fall 2025

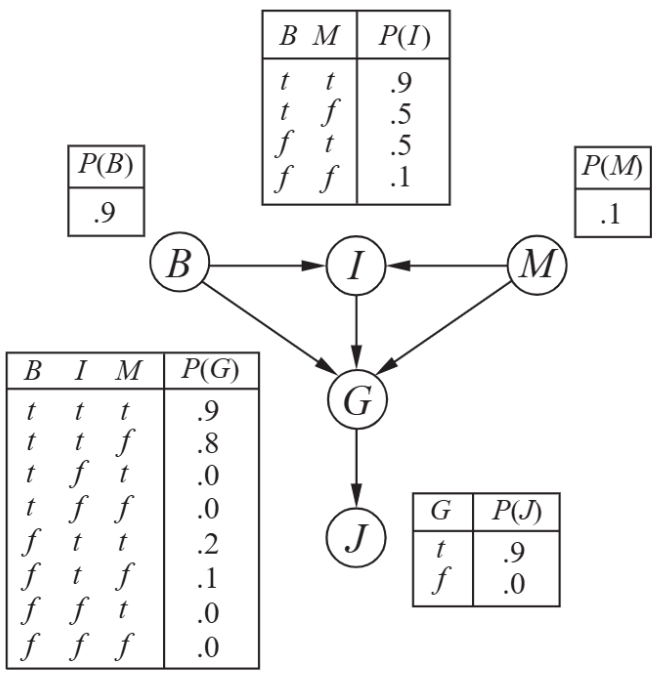
Problem Set 11 (45 pts)

Due at 11:59pm

Friday, November 21

Exercises are from *https://aimacode.github.io/aima-exercises/* after some fixes.

1.(14 pts) (Exercise 14.16) Consider the Bayes net below for five Boolean variables with meanings: $B = \{\text{BrokeElectionLaw}\}$, $I = \{\text{Indicted}\}$, $M = \{\text{PoliticallyMotivatedProsecutor}\}$, $G = \{\text{FoundGuilty}\}$, and $J = \{\text{Jailed}\}$.



$P(B)$
.9

$P(M)$
.1

B	M	$P(I)$
t	t	.9
t	f	.5
f	t	.5
f	f	.1

B	I	M	$P(G)$
t	t	t	.9
t	t	f	.8
t	f	t	.0
t	f	f	.0
f	t	t	.2
f	t	f	.1
f	f	t	.0
f	f	f	.0

G	$P(J)$
t	.9
f	.0

(a) (3 pts) Which of the following are asserted by the network structure?

$$\begin{aligned} \boldsymbol{P}(B, I, M) &= \boldsymbol{P}(B)\boldsymbol{P}(I)\boldsymbol{P}(M) & (1) \\ \boldsymbol{P}(J \mid G) &= \boldsymbol{P}(J \mid G, I) & (2) \\ \boldsymbol{P}(M \mid G, B, I) &= \boldsymbol{P}(M \mid G, B, I, J) & (3) \end{aligned}$$

(b) (4 pts) Calculate the probability $P(b, i, \neg m, g, j)$.

(c) (7 pts) Calculate the probability that someone goes to jail given that they broke the law, have been indicted, and face a politically motivated prosecutor.

2.(10 pts) (Exercise 14.18(a)) Consider the variable elimination algorithm given below. For the shown burglary-alarm network, applies variable elimination to the query

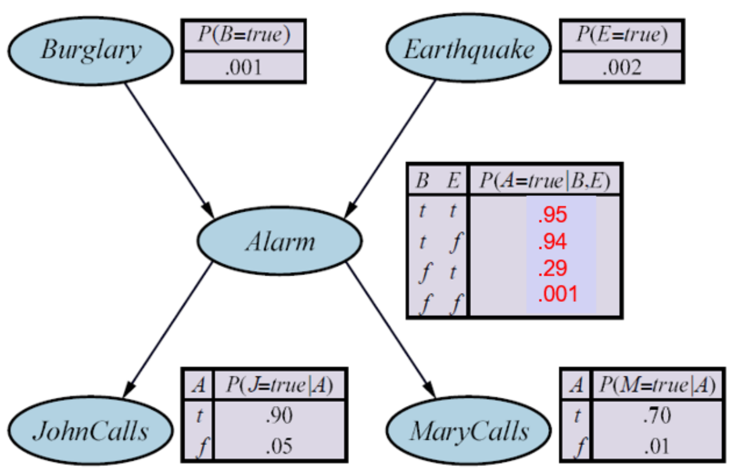
$$\boldsymbol{P}(Burglary \mid JohnCalls = True, MaryCalls = True).$$

Perform the calculations indicated and check that the answer is correct.

```
function ELIMINATION-ASK(X, e, bn) returns a distribution over X
  inputs: X, the query variable
          e, observed values for variables E
          bn, a Bayesian network specifying joint distribution P(X_1, ..., X_n)

  factors ← [ ]
  for each var in ORDER(bn.VARS) do
     factors ← [MAKE-FACTOR(var, e)|factors]
     if var is a hidden variable then factors ← SUM-OUT(var, factors)
  return NORMALIZE(POINTWISE-PRODUCT(factors))
```

Figure 14.11 The variable elimination algorithm for inference in Bayesian networks.

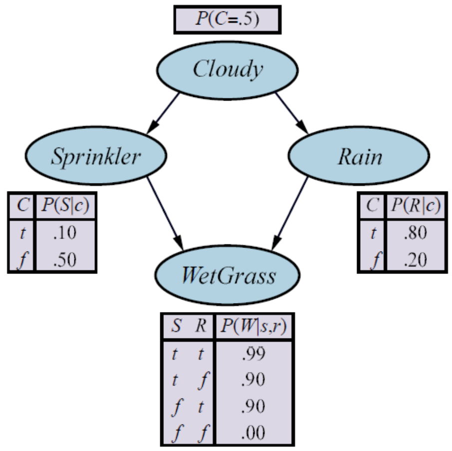


3.(21 pts) (Exercise 14.21) Consider the query $\boldsymbol{P}(Rain \mid Sprinkler = true, WetGrass = true)$ in the figure below and how Gibbs sampling via uniform choice of non-evidence variables can answer it.

(a) (2 pts) How many states does the Markov chain for this query have?

(b) (10 pts) Calculate the *transition matrix* Q containing the kernel $k(\boldsymbol{y} \to \boldsymbol{y}')$ for every directed edge from, say, a state $\boldsymbol{y}$ to another state $\boldsymbol{y}'$ or itself (in the case $\boldsymbol{y} = \boldsymbol{y}'$), in the Markov chain. Label the rows and columns of the matrix by the states such that an entry in the matrix corresponds to the probability of a transition from the state given by the row label to the state given by the column label.

(c) (2 pts) What does Q^2, the square of the transition matrix, represent?



(d) (3 pts) What about Q^n as $n \to \infty$?

(e) (4 pts) Explain how to do probabilistic inference in a Bayesian network, assuming that Q^n is available. Is this a practical way to do inference?